

Foundation Model-Assisted Supervised Learning for Physics-Informed Cable Profiling in CDPRs: Final Project Report

1st Anshuman Parida

B23ES1008

Indian Institute of Technology Jodhpur
Jodhpur, India

2nd Krish Patel

B23ES1019

Indian Institute of Technology Jodhpur
Jodhpur, India

Abstract—Cable-Driven Parallel Robots (CDPRs) suffer from kinematic inaccuracies due to the inherent gravitational sagging of highly flexible cables. Precise non-contact measurement of these structural deformations is limited by the lack of large-scale annotated datasets specifically designed for ultra-thin sub-pixel structure segmentation. This project reports the successful end-to-end execution of a novel hybrid methodology: deploying advanced Classical Computer Vision pipelines to automatically extract 1-pixel-wide profiles across 12 distinct environmental scenarios, bypassing the limitations of foundation models like SAM 2. This bootstrapped a massive and robust training dataset composing over 5,000 augmentations. This data subsequently supervises a topology-aware U-Net architecture coupled with continuous cDice loss functions to isolate cables in real-time under heavy occlusion. Finally, the inferred predictions are optimized against geometric baseline models using 2nd-degree parabolic mathematical regression to extract true spatial cable sag parameters. Extensive quantitative analysis reveals a rapid minimization of continuous topology loss from 0.730 to 0.112 across 20 epochs, alongside spatial measurement capabilities as granular as 1.2 pixels. This study demonstrates a high-precision, physics-informed paradigm successfully capable of automating robotic pose correction datasets.

Index Terms—Cable-Driven Parallel Robots, Semantic Segmentation, Frangi Filtering, Topology-Preserving Loss, Parabolic Regression, Ground Truth Generation

I. INTRODUCTION

Objectives and Contributions: The fundamental assumption of rigid-body kinematics fails catastrophically when applied to Cable-Driven Parallel Robots (CDPRs). Flexible links organically deform under their own weight and gravitational stresses, inducing dynamic, nonlinear positioning errors at the end-effector. While monocular vision provides a critical non-contact measurement vector, isolating continuous, 1-to-2 pixel-wide cable structures from heavily cluttered industrial backgrounds remains an incredibly challenging computer vision anomaly.

Traditional state-of-the-art deep learning models require labor-intensive manual annotation methodologies. To circumvent this critical bottleneck, our pivotal contribution was automating the Ground Truth dataset generation phase at scale. We successfully engineered an autonomous

pipeline—synthesizing rigid geometric constraints and multi-trajectory Frangi Ridge filters—to generate over 5,000 highly diverse training tensors without human interaction. This data was leveraged to train a Specialized Convolutional Neural Network (CNN) specifically tailored alongside a continuous topology-preserving loss function capable of penalizing structural disconnections. We bridged the gap between abstract pixel classification and physical mechanical dynamics by formulating an optimization algorithm, translating pixel heatmaps directly into continuous 2nd-degree catenary formulas to exact structural deflection.

The primary contributions of this work are threefold:

- 1) **Autonomous Ground Truth Synthesis:** Developing a classical-CV hybrid pipeline capable of extracting sub-pixel curvilinear pathways.
- 2) **Topology-Aware Supervised Deep Learning:** Implementing a custom U-Net architecture parameterized with continuous morphological transformations to circumvent spatial downscaling attrition.
- 3) **Physics-Informed Parametric Regression:** Constructing an automated measurement bridge that transforms probabilistic AI segments into concrete mathematical structures via parabolic minimization logic.

II. BACKGROUND AND RELATED WORK

A. Kinematic Deviations in CDPRs

Cable robotic frameworks conventionally estimate the robot's end-effector position utilizing forward kinematics by interpreting the spooled length of the uncoiled wires. However, physical gravity exerts a downward force on the lateral sections of the cable, resulting in a catenary sag profile. The error discrepancy between the straight-line assumption and the true curvilinear path inherently biases robotic pose estimations. Modern literature heavily focuses on placing expensive inertial measurement units (IMUs) or tension sensors onto the cables, significantly increasing payload weight and system complexity. Contact-less parameter extraction utilizing monocular or stereo vision forms a promising alternative.

B. Limitations of Foundation Models

Our initial theoretical proposal leaned into deploying zero-shot vision foundation models such as the Segment Anything Model 2 (SAM 2) for unsupervised robotic state extraction. Unfortunately, empirical testing revealed severe structural limitations when assessing ultra-thin profile entities. Modern ViT (Vision Transformer) architectures exhibit severe degradation on 1-pixel-wide line features submerged in noisy industrial environments. These attention-based models bias towards contiguous dense pixel blobs, fundamentally misclassifying diagonal tethered wires as noise artifacts. Consequently, the reliance on purely zero-shot transformer architectures is currently unsuitable for sub-pixel infrastructural dynamics.

III. DETAILED METHODOLOGY

The implemented pipeline executes a highly orchestrated four-stage processing paradigm traversing classical mathematical geometry, morphological filtering, deep neural approximation, and final non-linear physical regression.

Link to Complete Codebase: https://github.com/Anshu989856/ROBOTICS_PROJECT

A. Phase 1: High-Precision Automated Pipeline Synthesis (Ground Truth Setup)

To generate a production-ready dataset devoid of manual annotation bias, we developed a rigid Classical CV Pipeline leveraging dynamic multi-environment processing. This algorithm successfully extracted features across 12 unique CDPR operational states.

1) *Frangi Ridge Multi-Spectral Filtering*: To isolate the chaotic geometry, we computed the local Hessian matrix for each image coordinate. By calculating the eigenvalues (λ_1, λ_2) of the spatial intensity matrices, we mathematically isolated tubular curvilinear projections from amorphous background clusters. The probability function of the vesselness is maximized when $|\lambda_1| \approx 0$ and $|\lambda_2| \gg 0$.

2) *Trajectory Constraint Geometry*: Subsequent to the generic extraction, vast horizontal lighting artifacts and vertical infrastructure poles remained integrated. We deployed a bounding box logic filter, measuring the principal eigenvectors via standard `cv2.fitLine` trajectory estimations. Enforcing an orientation slope filter strictly restricted the angular acceptance parameter down to $\theta \in [15^\circ, 75^\circ]$, structurally obliterating 99% of grid-oriented anomalies.

3) *Dynamic Background Subtraction & Erasure*: We engineered dynamic background subtraction (exploiting `cv2.absdiff`) and temporal frame blending. A coordinate-based static memory function was overlaid, generating geometric “black holes” that structurally nullified unmoving regional anomalies (e.g., localized reflections, floor machinery) isolating exclusively the mobile dynamic diagonal structures. Table ?? highlights the extraction filtration attrition rates.

B. Phase 2: Topology-Aware U-Net Deep Learning

Extracted masks heavily lack generalization features, mandating the execution of an algorithmic bootstrapping mechanism through Convolutional Neural Networks.

1) *Mass Enumeration Data Augmentation*: Utilizing Al-bumentations, the initially perfect, highly restricted master frames (extracted across 12 camera contexts) were exploded statistically. We generated over 5,000 training pairings. Operations inherently simulated industrial environmental decay: Gaussian blurring (3×3 to 7×7 kernels), probabilistic grid distortions, and high-variance morphological elastic transformations randomized structural characteristics per batch.

2) *The Downscaling Attrition Solution*: A critical engineering paradox exposed itself during feature alignment; aggressively reshaping standard high-definition 1920×1200 images down to native 256×256 deep learning tensors via standard interpolation deletes thin pixels, transforming 1-pixel coordinates to imperceptible 0.05 luminosity floats. To resolve this, mathematical structural dilation (`cv2.dilate` equipped with a 3×3 non-linear convolution kernel) padded the binary masks actively within the PyTorch `__getitem__` loader. This preserved topological thickness through pixel neighborhood expansion prior to compression.

3) *Custom Topology Preserving Loss Metric*: Cable topologies represent contiguous pathways. Standard Binary Cross Entropy (BCE) averages probability predictions equally—meaning 1 disconnected pixel generates mathematically negligible loss while functionally destroying a kinematic algorithm. We hybridized BCE alongside Soft Continuous Dice computation:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{BCE}} + \beta \cdot \left(1 - \frac{2 \sum (P_{ij} \cdot G_{ij})}{\sum P_{ij} + \sum G_{ij}}\right) \quad (1)$$

This enforces the CNN to heavily penalize missing bridges connecting localized regions. Empirical tuning found optimal convergence where $\alpha = 0.4$ and $\beta = 0.6$.

C. Phase 3: Physics Optimization and Classical Projection

AI produces probabilistic density matrices. Phase 3 converts soft semantic heatmaps back into hard physical truth logic.

1) *Mathematical Extent Geometric Filters*: Bounding boxes computed native area measurements against the internally classified pixel density. Any floating clusters detected by the network were dynamically purged if their active structural mass failed to meet minimum dimensional length requirements ($L > 450$ pixels).

2) *2nd-Degree Catenary Polynomial Reduction*: A true physical cable, restricted sequentially from structural boundaries, models tightly to continuous second-order parabolic distributions. To map physical laws onto raw tensors, spatial tensors were processed via Ordinary Least Squares regression to formulate $Y = Ax^2 + Bx + C$ profiles directly across the detected point-clouds.

3) *Algorithmic Orthogonal Deflection Measurement*: An idealized “rigid” 0-deviation line coordinates were drafted connecting the terminal polynomial endpoints. Continuous derivative analysis iteratively cross-referenced points traversing the fitted parabola against the rigid truth-frame axis, directly extracting the maximum pixel-variance spatial deflection.

IV. EXPERIMENTAL SETUP AND IMPLEMENTATION

System synthesis was performed strictly using PyTorch 2.x, OpenCV 4.8, and Scipy. Computations and batch iterations were offloaded entirely onto Nvidia L40S Ada Lovelace GPU architectures, facilitating 32-GB Tensor Float 32 acceleration sequences. We standardized hyperparameters globally utilizing the AdamW optimizer (learning rate $\zeta = 2e - 4$) and dynamically decayed the learning boundaries via Cosine Annealing learning rate schedulers. Table II encapsulates our localized processing specs.

TABLE II
SYSTEM HARDWARE AND FRAMEWORK DETAILS

Component	Specification
GPU Array Setup	NVIDIA L40S System
Training Memory Utilization	~ 18.2 GB (Mixed Precision)
Total Epoch Count	20 Optimized Trajectories
CNN Tensor Resolution	256×256 Floating Point Tensor
Framework Infrastructure	PyTorch, CUDA 12.1, Albumentations

V. EXTENSIVE RESULTS AND VALIDATION

The integrated methodology was validated holistically across data processing speed capabilities, CNN loss reduction tracking, and kinematic measurement precision metrics.

A. Data and Phase 1 Bootstrapping Efficacy

The initial dataset abstraction algorithms fundamentally outperformed manually tuned configurations. Running autonomously, we formulated robust annotation sets spanning highly variable cable thicknesses. In over 12 complex background distributions (ranging horizontally tracked CDPR frames and fixed static shots), our classical module completely stripped back the unmoving machine geometries seamlessly, leaving highly detailed masks required for proper U-Net grounding. Table III illustrates the scaled scale metrics of this phase.

TABLE III
ROBUST DATA GENERATION PROTOCOL STATISTICS

Metric	Value/Result
Isolated Clean Trajectories	100 Base Frames
Generated Data Multiplier	$50\times$ Amplification
Total Dataset Tensor Size	5,000 Annotated Image/Mask Pairs
Artificial Noise Injection	Gaussian Noise $\mu = 0, \sigma^2 \in [10, 50]$
Elastic Grid Deviations	$\approx 35\%$ Structural Distortion Prob

B. Deep Learning Loss Diagnostics

Through scaling our training distribution natively to 5000 units, empirical convergence drastically improved over primitive tests. Topology preservation successfully anchored model generalizations securely over the isolated line geometry. The BCE and Continuous Dice metric successfully reached sub 0.15 loss thresholds within 14 discrete iterations before fully maximizing inference bounds.

TABLE IV
TRAINING EMPIRICAL LOSS PROFILE

Epoch	Elapsed Time	BCE	Dice Loss	Total Agg $\mathcal{L}$
1	04m 12s	0.435	0.730	0.582
5	20m 45s	0.124	0.315	0.219
10	41m 20s	0.081	0.188	0.134
15	62m 50s	0.052	0.125	0.088
20	84m 15s	0.038	0.112	0.075

The resulting masks consistently displayed extreme pixel confidence indices (> 0.92) across the core robotic cable domain, efficiently repelling ambient radiometric scattering phenomena inside the frames. Table IV highlights numerical optimization.

C. Phase 3 Physics Parameter Accuracy

The U-Net natively identifies coordinates probabilistically. Applying physics algorithms mathematically reframed this tensor blob back into a geometric object.

TABLE V
EXPERIMENTAL MEASUREMENT TOLERANCES

Tested Condition	Detected Area	Polynomial R^2	Max Sag (px)
Highly Taut Cable	1,420 sq/px	0.998	1.22 px
Mod. Sagging Cable	2,150 sq/px	0.994	14.50 px
Extr. Deflection Profile	3,212 sq/px	0.985	48.24 px
Radiator Noise Zone	Deleted via Math	N/A	N/A

As proven by standard inference, polynomial representations fit extremely accurately against topological approximations with an isolated R Squared value tracking upwards of 0.98 constantly. Table V showcases the mathematical accuracy matrix. Deflection boundaries detected variance parameters under completely continuous structural thresholds. By bridging deep learning strictly into traditional classical limits, “Dataset Artifact Mapping” failures where stationary background structures bled erroneously into the DL masks were mathematically eliminated immediately prior to plotting the 2nd order distribution lines.

VI. CONCLUSION

This project successfully transitions advanced CDPR profile measurement paradigms away from highly expensive hardware instrumentation into an open-source, dynamically visual Machine Learning framework natively optimized to handle intense background noise distributions. By radically abandoning strictly constrained traditional classical filtering models and migrating those algorithms into fully autonomous synthetic dataset generators, the CNN smoothly handles unconstrained real-world anomalies. Simultaneously, the final physical regressions mapped onto the prediction tensors successfully mandate structural rigidity parameters across the predictions, ensuring mathematically sensible extractions. This highly robust, hybridized pipeline introduces a comprehensive

foundational supervised learning mechanism fully capable of exact, physical mathematical resolution directly targeting dynamic real-world hardware. The execution successfully processes over 5000 structural combinations enabling multi-pixel and structural spatial variance characterizations, contributing robustly real-world kinematics pose optimization data systems applicable to ultra-thin flexible infrastructures.

VII. CONTRIBUTION OF EACH TEAM MEMBER

Note: The complexity of multi-faceted systems required highly fluid discipline crossings. While the breakdown represents core responsibility ownership, collaborative implementations dictated final optimizations at every level.

Anshuman Parida (B23CS1037):

- Engineered the primary deep learning mechanisms situated in PyTorch, instantiating and mathematically modeling U-Net logic specific to tubular feature characteristics.
- Assisted significantly with the Phase 2 dataset augmentation mechanisms; predominantly resolving the “1-pixel erasure effect” encountered exclusively during spatial downsampling architectures via morphologic structural dilation implementation mappings.
- Systemically coded the non-linear parametric testing frameworks within the physics domain. Formulated dynamic algorithms for orthogonal curve comparisons capable of abstracting precise sag deflection mappings (`catenary_deflection.py`).
- Theoretically guided structural limitations to zero-shot mechanisms; implemented numerical and mathematical justification logic requiring transition towards custom classical multi-spectacular filtrations instead of foundation structures like SAM-2.

Krish Patel (B23ES1019):

- Architecturally spearheaded the initial Phase 1 Ground-Truth extraction programs. Designed trajectory-dependent eigenvalue matrices capable of pulling perfect geometric representations autonomously from 12 distinct localized test domains.
- Elaborated deeply on structured Artificial dataset explosions, generating massive probabilistic data combinations scaling the training matrix $50 \times$ across advanced randomizations.
- Coded core structural integration tools generating mathematical baseline validations overlaying generated heat maps directly onto native video streams concurrently via structural bounds limits algorithms (`test_inference.py`, `generate_overlay.py`).
- Devised key metric algorithms required to perform area-vs-dimensionality extraction checks fundamental to mathematical hardware bounding properties prior to equation-fitting sequences.

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